

Ames Housing Exploratory Data Analysis

End-to-End Data Analysis Capstone Report

1. Project Overview

Primary question: What property characteristics are most strongly associated with residential sale prices in Ames, and what unusual observations or data-quality issues should be considered when interpreting those relationships? The analysis follows the notebook pipeline: raw-data assessment, missing-data analysis, cleaning, univariate and bivariate EDA, correlation analysis, outlier investigation, multivariate analysis, hypothesis testing, findings, and limitations. The work uses 1,460 property records and 11 variables, with SalePrice as the target, Neighborhood as a categorical predictor, numerical/ordinal predictors, and Id as an identifier.

2. Data Quality and Cleaning

Initial checks found 248 missing LotFrontage values and no missing SalePrice values. There were 0 duplicate rows and 0 duplicate Id values. Domain checks found no invalid values for OverallQual, GrLivArea, YearBuilt, Bedrooms, FullBath, GarageCars, LotArea, SalePrice, or observed LotFrontage under the notebook's stated rules.

Metric	Result
Raw dataset	1,460 rows × 11 columns
Missing LotFrontage	248 (17.0%)
SalePrice mean / median	\$237,645 / \$230,600
OverallQual ↔ SalePrice Pearson r	0.799
GrLivArea Pearson r / Spearman ρ	0.601 / 0.611
OverallQual test Spearman ρ	0.7919
Neighborhood H / ε²	1142.9893 / 0.7813

The notebook compares dropping missing rows, overall-median imputation, and neighborhood-specific median imputation. Dropping retains 1,212 observed values, while median-based approaches retain all 1,460 rows. The observed LotFrontage mean and standard deviation are 70.02 and 21.90; neighborhood-median imputation produces 69.90 and 19.98. The final analysis uses neighborhood-median imputation, with an overall-median fallback if required.

3. Exploratory Findings

SalePrice has a mean of \$237,645, median of \$230,600, standard deviation of \$65,093, and ranges from \$92,100 to \$595,000. Its skewness is 0.667, indicating a right-tailed distribution. OverallQual has the strongest numerical correlation with SalePrice ($r = 0.799$), followed by GrLivArea (0.601), FullBath (0.396), GarageCars (0.326), YearBuilt (0.272), Bedrooms (0.254), and LotArea (0.101). LotFrontage has a near-zero correlation (-0.011). Correlation describes association rather than causation.

The GrLivArea/SalePrice relationship is positive and broadly monotonic (Pearson $r = 0.601$; Spearman $\rho = 0.611$), although the scatter contains deviations and extreme cases.

Neighborhood medians are highest for NoRidge (\$373,000), NridgHt (\$367,700), and StoneBr (\$363,300); lower observed medians include MeadowV (\$167,450), IDOTRR (\$169,700), and BrDale (\$175,100). These are descriptive differences, not causal neighborhood effects.

4. Quality, Size, and Outliers

Quality bands show a descriptive price gradient: median SalePrice is \$179,350 for Low (1–4), \$235,600 for Medium (5–6), \$296,900 for High (7–8), and \$371,550 for Very high (9–10). The multivariate analysis examines the size-price relationship across quality groups and combines Neighborhood and QualityBand using a median-price heatmap.

The IQR procedure identifies 8 potential GrLivArea outliers, 11 LotArea outliers, and 6 SalePrice outliers. Two large homes are Id 1016 in Timber (4,600 sq ft, OverallQual 5, \$160,000) and Id 858 in IDOTRR (4,500 sq ft, OverallQual 6, \$145,000). Both observations passed the notebook's stated domain-validity checks, and no apparent data-entry errors were identified during this review. They were retained rather than automatically deleted and are investigated as the unexpected finding in Section 6.

5. Hypothesis Testing

Test 1 — OverallQual and SalePrice. Spearman rank correlation is used because OverallQual is ordinal/discrete and a monotonic relationship is of interest. The result is $p = 0.7919$. The notebook displays $p = 0.0000e+00$ because the value is below its numerical display range, not because the true p-value is zero. The effect size is the same p value. Interpretation should consider effect size and diagnostic plots, not the p-value alone.

Test 2 — Neighborhood and SalePrice. The Kruskal-Wallis H test is used for multiple independent neighborhood groups. The result is $H = 1142.9893$, $p = 3.11 \times 10^{-235}$, with epsilon-squared = 0.7813. This is consistent with differences in sale-price distributions across neighborhoods; it does not establish causation.

6. Multivariate Interpretation and Key Takeaways

SalePrice is associated with several property characteristics, with OverallQual and GrLivArea standing out among numeric predictors. Neighborhood distributions also differ, and the Neighborhood $\times$ QualityBand analysis shows that price patterns depend on more than one characteristic. The size-price relationship should therefore be interpreted in the context of property quality.

LotFrontage missingness is concentrated unevenly across neighborhoods. The notebook compares imputation strategies before selecting a neighborhood-aware approach, preserving the full sample while accounting for neighborhood context.

Unexpected finding — The largest houses were not among the most expensive. Id 1016 in Timber had 4,600 sq ft and sold for \$160,000, while Id 858 in IDOTRR had 4,500 sq ft and sold for \$145,000. Their OverallQual scores were 5 and 6, respectively. Although GrLivArea has a positive correlation with SalePrice ($r = 0.601$), these observations show that living area alone does not determine property value. The quality-banded analysis in notebook Section 20 examines how this relationship differs across quality groups. Both properties passed the notebook's stated domain-validity checks, and no apparent data-entry errors were identified during this review. This highlights the importance of considering quality alongside living area.

7. Limitations

The analysis is observational, so associations should not be interpreted as causal effects. LotFrontage required imputation, and the chosen method can affect its distribution. Extreme observations can influence correlations and summary statistics even when legitimate. Neighborhood group sizes differ, affecting distribution comparisons. Findings from this Ames dataset may not generalize to other cities, time periods, or housing markets.

8. Conclusion

The completed EDA provides a reproducible view of factors associated with Ames residential sale prices. Overall quality shows the strongest numeric association with SalePrice, living area has a substantial positive association, and sale-price distributions differ across neighborhoods. The project combines data-quality checks, transparent missing-data handling, purposeful visualizations, outlier review, multivariate analysis, and effect-size-aware hypothesis testing while distinguishing association from causation.

Reproducibility: The analysis is implemented in Ames_Housing_EDA_Capstone.ipynb. Final validation confirms 1,460 raw rows, 11 columns, unique Id values, non-missing SalePrice, and zero remaining LotFrontage values after cleaning.